

VISHAL SONI

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Education

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering | CGPA: 9.01/10

Sep 2023 – Present

Bhopal, India

Experience

Indian Institute of Technology Ropar — VLED Lab

May 2026 – Present

Software Engineering Intern

Remote

- Contributing to open-source educational infrastructure tools under IIT Ropar faculty mentorship; involved across design, implementation, review, and deployment.

Open Source

- eSim (FOSSEE, IIT Bombay)** — Ported eSim EDA to Ubuntu 25.04 by resolving 13 dependency and toolchain failures across KiCad, GHDL, LLVM, Python 3.13, and package repositories.

Projects

CollabCode — Real-Time Collaborative IDE (*Next.js, TypeScript, Go, WebSocket, PostgreSQL*) Feb – May 2026

- Engineered a low-latency collaborative coding platform with a custom Operational Transform engine to resolve concurrent edit conflicts across distributed sessions.
- Designed a Pub/Sub scaling architecture in Go Fiber handling multi-room WebSocket sessions with JWT-authenticated isolation; integrated Monaco Editor with live code execution.
- Built multi-file workspace management, room-scoped chat, and a structured REST API layer backed by PostgreSQL with session-aware state persistence.

ChronoVault — Distributed Encrypted File Vault (*Go, React, FastAPI, AES-256-GCM, IPFS*) Nov 2025 – Feb 2026

- Designed the core encryption and storage layer: AES-256-GCM encryption, deterministic file sharding, and Merkle-tree integrity verification for tamper-evident chunk storage.
- Implemented decentralized-style chunk distribution over IPFS with secure file reconstruction, biometric authentication, and cryptographic key derivation workflows.
- Collaborated within a 5-member team, contributing to system architecture decisions, cross-component integration, and technical design across the full stack.

Arvis X — AI-Powered Travel & Booking Platform (*Next.js, Cashfree, AWS Lambda, Kubernetes*) Sep 2024 – May 2025

- Built a full-stack travel and booking platform supporting buses, trains, flights, hotels, and events with role-based dashboards for users, vendors, and administrators.
- Engineered real-time seat reservation using Redis locking, WebSockets, and distributed cache invalidation; integrated Cashfree payments, wallet, coupon engine, QR e-tickets, and automated refunds.
- Developed AI-powered travel assistant, itinerary generation, and personalized recommendations using LLM APIs; deployed on AWS Lambda and Kubernetes with Terraform-managed IaC.

DineOps — Restaurant Management Ecosystem (*Next.js, Cashfree TypeScript, Prisma, Supabase*) Jun 2024 – Sep 2024

- Developed a production-grade restaurant management platform with customer, cashier, kitchen, delivery, admin, and super-admin portals using role-based access control.
- Built end-to-end order workflows: POS operations, kitchen display systems, delivery tracking, reservations, inventory management, and AI-powered concierge for personalized dining.
- Designed dual-mode persistence with PostgreSQL production storage and offline JSON fallback; deployed on Vercel and Supabase with JWT auth, real-time notifications, and CI/CD pipelines.

Achievements

- AIR 217** in TCS CodeVita Season 13 Round 1 (100,000+ participants); ranked **3rd at VIT Bhopal** university-wide; final global rank **4,771**.
- Finalist, Smart Delhi Hackathon** — selected from 15,000+ participants for city-scale solution development track.
- Selected as mentee (**top 0.1%** of 80,000+ applicants) for **Codeforces Graphs Camp**; mentored by a Codeforces Master on hard graph problems (shortest paths, traversal, connectivity).

Technical Skills

Languages: Go, Python, C++, Java, TypeScript/JavaScript, PHP, SQL

Backend & Systems: REST APIs, WebSockets, Operational Transform, Serverless, PostgreSQL, MySQL, DynamoDB, Redis

Cloud & Infrastructure: AWS (Lambda, EC2, EKS, RDS, S3, ALB, IAM), GCP (Cloud Run, Firebase), Kubernetes, Terraform

DevOps & Observability: Docker, GitHub Actions, Jenkins, Prometheus, Grafana

Tools & OS: Linux, Git, IPFS, KiCad/eSim, LLVM toolchains